

Home work 1

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1 Task

$$A = \begin{bmatrix} 141 & 121 & 161 \\ 262 & 232 & 292 \end{bmatrix}$$

$$B = \begin{bmatrix} 4 & -2 \\ -2 & 1 \\ -6 & 3 \end{bmatrix}$$

$$C = \begin{bmatrix} 1 & 0 & -4 \\ 2 & 0 & -8 \end{bmatrix}$$

Find: $(AB)^2C = ?$

$$\begin{aligned} (AB)^2C &= AB * A * (B * C) \\ (B * C) &= \begin{bmatrix} 4 * 1 - 2 * 2 & 0 & 4 * (-4) - 2 * (-8) \\ -2 * 1 + 1 * 2 & 0 & -2 * (-4) - 8 \\ -6 * 1 + 3 * 2 & 0 & -6 * (-4) - 3 * 8 \end{bmatrix} = 0 \end{aligned}$$

Answer: $(AB)^2C = 0$

2 Task

$$A = \begin{pmatrix} 12 & 23 & 47 \\ -5 & 31 & -6 \\ 35 & 51 & -3 \end{pmatrix}$$

$$B = \begin{pmatrix} 3 & 0 & -3 \\ -4 & 2 & 7 \\ 1 & -2 & -4 \end{pmatrix}$$

$$C = \begin{pmatrix} -2 \\ 3 \\ -2 \end{pmatrix}$$

Find: $A^2B^3C = ?$

$$B^3C = (3 * (-2) - 3 * (-2) \quad -4 * (-2) + 2 * 3 + 7 * (-2) \quad -2 - 2 * 3 - 4 * (-2))$$

$$= (0 \quad 0 \quad 0)$$

Answer: $A^2B^3C = 0$

3 Task

$$A = \begin{pmatrix} 10 & -4 \\ 11 & -4 \\ -3 & 1 \\ -2 & 1 \end{pmatrix}$$

$$B = \begin{pmatrix} -5 & 7 & 11 & -3 \\ -3 & 11 & 27 & 5 \end{pmatrix}$$

Find: Let $C = AB$. $f(C) = ?$, where $f(x) = x^2(x + I)$ and $I = 1$ matrix.

$$f(C) = x^3 + x^2$$

$$= AB * AB * AB + AB * AB = A * (BA) * BAB + A * (BA) * B$$

$$BA = \begin{pmatrix} -5 * 10 + 7 * 11 + 11 * (-3) - 3 * (-2) & -5 * (-4) + 7 * (-4) + 11 - 3 \\ 0 & 0 \end{pmatrix}$$

$$= 0$$

Answer: $f(C) = x^3 + x^2 = 0 + 0 = 0$

4 Task

Let $x \in R^{n \times 1}$ and $y \in R^{n \times 1}$, and $A = xy^T$. Demonstrate, that you can find constant $\lambda \in R$, for any $k \in N$ performs $A^k = \lambda^{k-1}A$.

5 Task

Let V be a linear real vector space. Recall that, by the definition of a linear space, there exists a vector $\theta \in V$ such that $x + \theta = \theta + x = x$ for any $x \in V$. Using the axioms of a linear real space, prove that $0 \cdot x = \theta$.

Hint: Use the property that for each x there exists a unique $-x$ such that $x + (-x) = \theta$, and $(\alpha + \beta) \cdot x = \alpha \cdot x + \beta \cdot x$.

let's prove that $0 * x = \theta$

- 1) Property of the additive identity element (zero) in the field of real numbers
 $0 = 0 + 0$
- 2) Axiom of distributivity of multiplication over scalar addition (longread)
 $(\alpha + \beta) \cdot x = \alpha \cdot x + \beta \cdot x$ means
 $0 * x = (0 + 0) * x = 0 * x + 0 * x$
- 3) let $y = (0 * x)$

got $y = y + y$

4) add $-y$ to both sides from 3

got $y - y = y + y - y$

5) Axiom of associativity of addition (applied to the right side)

left side $y - y = \theta$

$\theta = y + y - y$

6) because $y - y = \theta$

got $y + \theta = y$

then got $\theta = y + \theta = y$

Transitivity of equality

got $y = \theta$

got $0 * x = \theta$

Result: we approved what we need